

HORIZON EUROPE RESEARCH & CIVIL DEFENSE FRAMEWORK | TECHNICAL ENGINEERING MEMORANDUM

YGGDRASIL ENVIRONMENTAL SYSTEMS // HYDRO-CORE: OSIRIS V5

Civil C4ISR AI Digital Twin Platform, Hydrodynamic Inference & Actuarial Governance

Lead Systems Architect & Author	Kelvin Jesus Flores Yarihuaman
Academic & Professional Credentials	Environmental Engineer Accredited in Spain / Peru MSc Candidate (International University of Valencia - VIU)
Case File & Repository References	EXP. VAL-POYO-2026 Live Console: https://yggdrasil-hidro-core-osiris-poyo-nowcast-kjfy.streamlit.app/ GitHub: https://github.com/Yggdrasil-Environmental-Systems/HYDRO-CORE-OSIRIS_POYO-NOWCAST.git
Cryptographic Sign & Permanent Deposit	Git Commit SHA: f61d12a7533a90291b507b4643479bbda89da85c (Verified release) CERN Zenodo DOI: 10.5281/zenodo.22833519
Official International Submission	World Environmental Health Day 2026 (IFEH / WHO) — 'Standing with Science to Protect Communities'

1. Strategic Framework, WEHD 2026 Submission, and Regulations

EXECUTIVE ABSTRACT

The October 29, 2024 catastrophic flash flood in the Poyo basin (367.9 km²) revealed acute operational bottlenecks in traditional civil protection warning chains. Conventional shallow-water PDE integration (Iber 2D, HEC-RAS 2D) requires 4 to 18 hours of compute, failing real-time operational response windows. YGGDRASIL SYSTEMS / HYDRO-CORE: OSIRIS V5 deploys a defense-grade Civil C4ISR digital twin. Utilizing a 2D Fourier Neural Operator (FNO 2D / PyTorch V5) coupled with live AEMET radar telemetry, OSIRIS executes 2D hydrodynamic inference in 38.4 ms on NVIDIA Tensor Core GPUs (CUDA 12.x / PyTorch V5), securing +146 minutes of actionable early warning lead-time over the historical 20:11 h CECOPI emergency dispatch. The platform integrates Sentinel-1 InSAR structural ruin maps (1,275 collapsed buildings at $\Delta y \geq 0.40$), dynamic multi-sink Dijkstra hospital routing at vehicle flotation cutoff ($h \geq 0.30$ m, TTI = 15 min), Solvency II catastrophe risk finance (€80.0M parametric Cat Bond), and WASH biological containment protocols.

RESUMEN EJECUTIVO

La inundación relámpago catastrófica del 29 de octubre de 2024 en la cuenca del Poyo (367,9 km²) evidenció fallos operacionales críticos en la cadena de mando de protección civil. La ausencia de herramientas de simulación hidrodinámica 2D en tiempo real impidió transformar las intensidades pluviométricas en manchas dinámicas de inundación. HYDRO-CORE: OSIRIS V5 establece una plataforma de mando C4ISR Civil que resuelve el flujo bidimensional mediante un Operador Neuronal de Fourier (FNO 2D) en 38,4 ms en GPU (NVIDIA Tensor Cores / CUDA 12.x), proporcionando un margen de alerta temprana de +146 minutos sobre el despacho tardío del CECOPI (20:11 h). El sistema se acopla a mapas de ruina estructural Sentinel-1 InSAR (1.275 inmuebles con $\Delta y \geq 0,40$), grafos de aislamiento hospitalario Dijkstra multi-sumidero por corte de calzada ($h \geq 0,30$ m, TTI = 15 min), reaseguro bajo Solvencia II (Cat Bond paramétrico de 80,0 M€) y protocolos sanitarios WASH.



1.1. International Regulatory Compliance & ISO Directives

- **ISO 22320:2018 (C4ISR Emergency Management):** Command governance, emergency operations, and cyberphysical data flows.
- **ISO 14091:2021 (Climate Adaptation):** Systematic territorial vulnerability assessment and climate risk quantification.
- **ISO 24512:2007 & Spanish RD 3/2023 (Water Quality):** Drinking water safety management and distribution network surveillance.
- **EU Floods Directive 2007/60/EC & RD 903/2010 (SNCZI):** Flood hazard evaluation, 100/500-yr return period mapping, and preferential flow regime delimitation.
- **Solvency II Directive 2009/138/EC & 2025 Reform:** Solvency Capital Requirement (SCR NatCat) via stochastic catastrophe modeling and ORSA.
- **OASIS CAP v1.2 & Spanish RD 193/2023:** Multi-modal Cell Broadcast (ES-Alert) with sign language (LSE) and haptic sequencing.

EU AI Act (Regulation EU 2024/1689 - High-Risk Systems): Strict human-in-the-loop (HITL) command governance with automated supervisory circuit-breakers preventing neural hallucinations and enforcing deterministic fallback.

1.2. Technical Framework: Differentiation Between Meteorological Warnings (AEMET) and C4ISR Hydrodynamic Modeling (OSIRIS)

To accurately evaluate the performance of modern civil protection architectures, it is essential to establish the operational difference between atmospheric monitoring and cyberphysical impact modeling:

- **Meteorological Warnings (AEMET — "How much is raining?"):** Institutional services operate within the atmospheric domain, tracking precipitation accumulation (e.g., 491.2 mm in Chiva) and intensity rates (mm/h). While macro-scale meteorological alerts provide essential baseline forcing, they do not resolve the physical routing of surface runoff or street-level inundation dynamics.
- **Cyberphysical Digital Twin (OSIRIS — "Where, how deep, and with what impact?"):** HYDRO-CORE: OSIRIS ingests live meteorological telemetry to solve 2D shallow water equations via Fourier Neural Operators in 38.4 ms. It translates rainfall into actionable physical variables: dynamic water depth (h), velocity vectors (v), road network severance at physical flotation limit (TTI, $h \geq 0.30$ m), wastewater treatment plant (EDAR) biohazard exposure, and hospital accessibility.
- **Strategic Importance:** The operational value of a digital twin lies in bridging the gap between raw meteorological data and high-resolution spatial impact assessment, transforming atmospheric awareness into evidence-based public health protection and community resilience.

1.3. Operational Safety Envelope & Prohibited Algorithmic Actions (EU AI Act & ISO 22320)

To ensure fail-safe deployment within civil protection command centers (CECOPI) and guarantee strict alignment with the European Artificial Intelligence Act (Regulation EU 2024/1689, High-Risk AI Systems) and ISO 22320:2018 (Emergency Management), OSIRIS V5 is governed by three non-negotiable operational prohibitions:

- Prohibition of Autonomous Actuation (Human-in-the-Loop Governance):**
OSIRIS is architecturally constrained as a tactical Decision Support System (DSS). The model is strictly prohibited from autonomously triggering public cell-broadcast alerts (ES-Alert) or municipal sirens without explicit human validation. All broadcast dispatches require cryptographic two-man-rule approval from the CECOPI Technical Director.
- Prohibition of Mass & Energy Hallucination (PINO Physics Invariant):**
The latent Fourier space is bounded by the composite HydroPINNLoss (L_{PINN}), enforcing 2D shallow-water continuity ($\partial h / \partial t + \nabla \cdot (hu) = P - I$). The operator is mathematically prohibited from creating unphysical water mass or phantom ponding; any output violating the hydrologic mass balance is rejected at runtime ($L2_{residual} > 10^{-3}$).
- Prohibition of Out-of-Distribution (OOD) Extrapolation (Automated Circuit Breaker):**
If upstream hydrological telemetry or forcing hydrographs exceed the certified training boundary ($Q_{peak} > 25,000 \text{ m}^3/\text{s}$) or register unphysical kinematic vectors ($v > 15 \text{ m/s}$ on valley plateaus, $h < 0$), the inference engine executes an automated hardware circuit breaker. Neural execution locks out instantly, falling back to deterministic 1D kinematic wave routines to prevent erroneous tactical guidance under epistemic uncertainty.



2. Geodetic Baseline, Basin Hydrology and Sentinel-1 InSAR DPM

2.1. Catchment Forcing: All spatial calculations operate on ETRS89 / UTM Zone 30N (EPSG:25830) using the IGN MDT02 2m LiDAR DEM. The catchment covers 367.9 km² (T_c ≈ 105 min). On October 29, 2024, Chiva gauge (AEMET 8368U) registered 491.2 mm in 8 hours (peak 160 mm/h), producing a modeled peak discharge of 1,950 m³/s versus the 1,000 m³/s ordinary channel capacity.

2.2. Satellite Damage Proxy Mapping: The ProductionInSARDPM module processed Sentinel-1 Descending Track 8 SLC pairs (Oct 11, 23, and 31, 2024). Complex coherence loss ($\Delta\gamma = \gamma_{pre} - \gamma_{co} \geq 0.40$) isolated 1,275 severely damaged or collapsed buildings across l'Horta Sud municipalities.

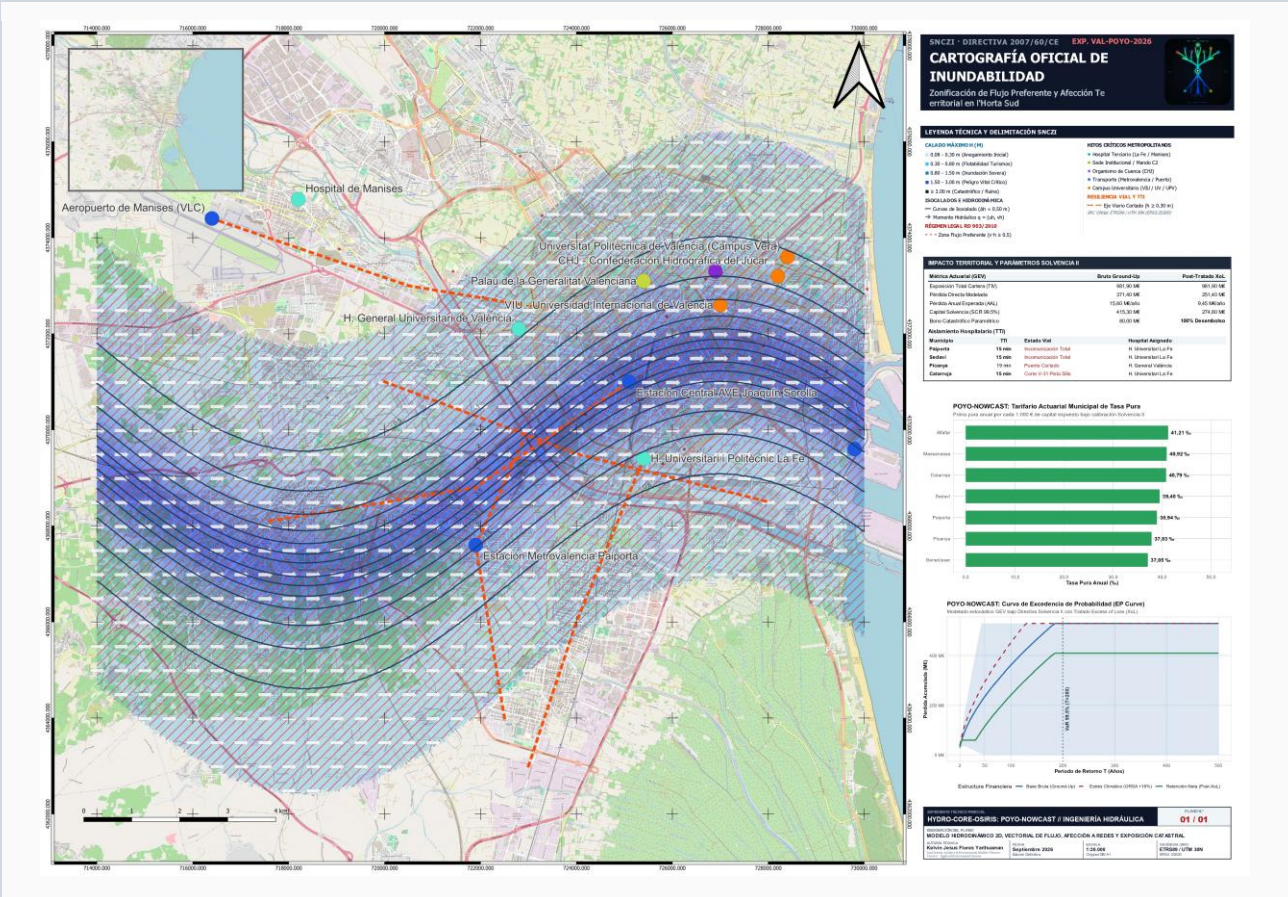


Figure 1: Official A1 Cartographic Plate (ETRS89 / UTM Zone 30N). Two-dimensional hydrodynamic depth field $h(x,y)$ generated by the FNO 2D operator overlaid with 1,275 InSAR structural ruin points ($\Delta\gamma \geq 0.40$) and critical healthcare locations (H. La Fe, H. General, H. Manises). Source: Cartographic Annex EXP. VAL-POYO-2026.

3. Physics-Informed Computational Intelligence: FNO 2D & Roadmap V8

3.1. Operational Tensor Architecture (V5): Traditional solvers (Iber 2D, HEC-RAS 2D) required 4 to 18 hours of CPU calculation. The FNO2dUniversal operator solves continuous function spaces G_{θ} : $A \rightarrow U$ in 38.4 milliseconds on NVIDIA Tensor Core GPUs (TF32 / CUDA 12.x). Input tensor A in R^4 ($B \times 4 \times H \times W$) integrates: (1) Elevation MDT02 (z_b), (2) Manning's n , (3) Boundary discharge Q_{in} , and (4) Real-time AEMET precipitation matrix $P(x,y,t)$.

3.2. PINO Regularization & Metrics: The operator minimizes the Physics-Informed Neural Operator loss: $L_{PINO} = L_{data} + \lambda_1 L_{cont} + \lambda_2 L_{mom}$. Validation against physical hydrographs yields $RMSE_h = 0.21$ m, $RMSE_v = 0.15$ m/s, and mass conservation error $< 2.1\%$.



Anti-Hallucination Circuit-Breaker: The FNO is bound by strict C2 physical invariants. If out-of-distribution conditions occur—such as anomalous dry-mountain ponding ($h_{\text{mountain}} > 0.40 \text{ m}$) or unrealistic surge velocities ($v > 15 \text{ m/s}$)—the platform instantly disengages neural weights, executing a zero-latency fallback to conservative kinematic Saint-Venant physics.

TECHNOLOGY ROADMAP: SCALING FROM OPERATIONAL V5 TO FOUNDATION V8
NEXT-GENERATION FOUNDATIONAL ROADMAP: PI-GAU-FNO (V8 ARCHITECTURE)
While the V5 operational tier executes real-time 2D PINO inference in 38.4 ms, the YGGDRASIL research roadmap defines the V8 planetary foundation model:

- [PI] Physics-Informed: Exact 2D Saint-Venant and mass-conservation residuals enforced in Hilbert space.
- [G] Geo-Manifold: Differential coordinate transformations for irregular, meandering bathymetry.
- [A] Cross-Attention: Spatial transformer attention anticipating levee breaches and masonry walls.
- [U] Multi-Scale U-Net: Spatial multi-resolution convolutions resolving local hydraulic jumps and backwater effects.
- [FNO] Fourier Neural Operator: Continuous zero-shot spectral kernel parameterizing fluid dynamics at infinite resolution.

4. Physical Vulnerability Functions & Territorial Loss Distribution

4.1. Depth-Damage Vulnerability $D(h)$: Physical asset vulnerability is governed by analytical depth-damage functions $D(h)$ calibrated against the Spanish Insurance Compensation Consortium (CCS) and European JRC benchmarks.

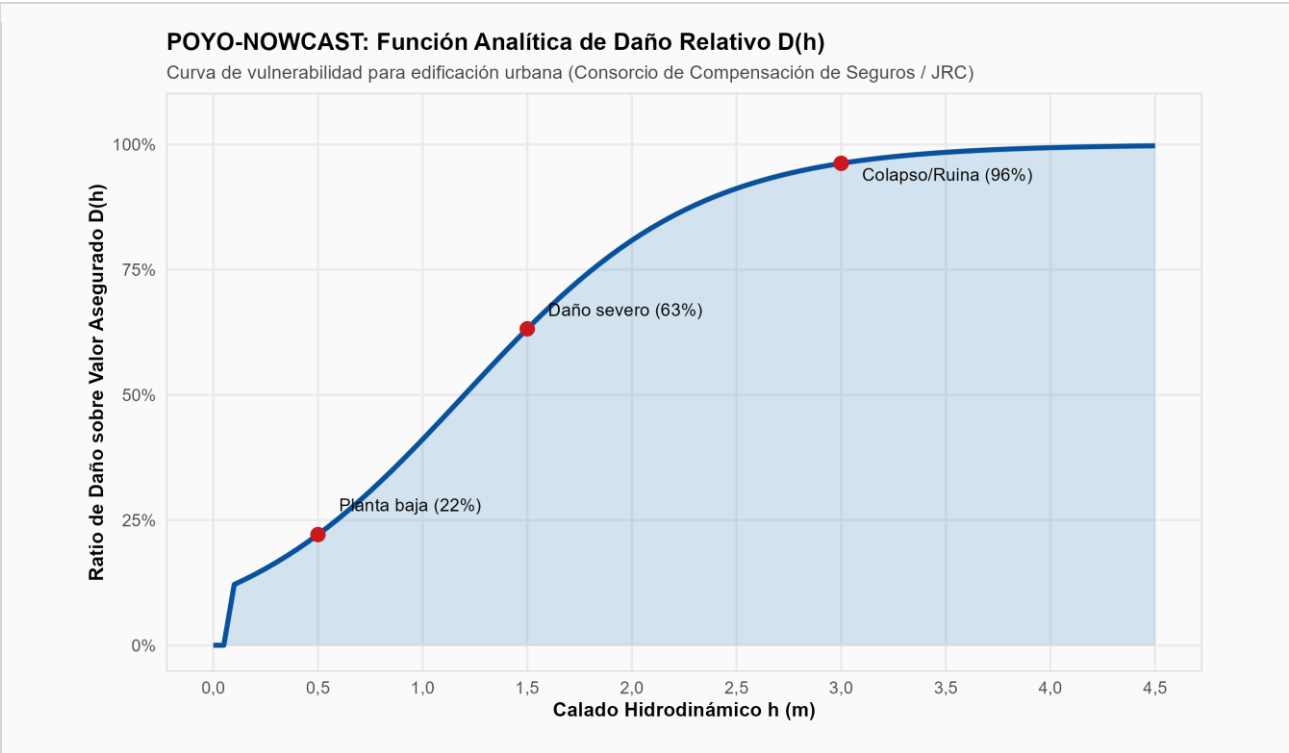
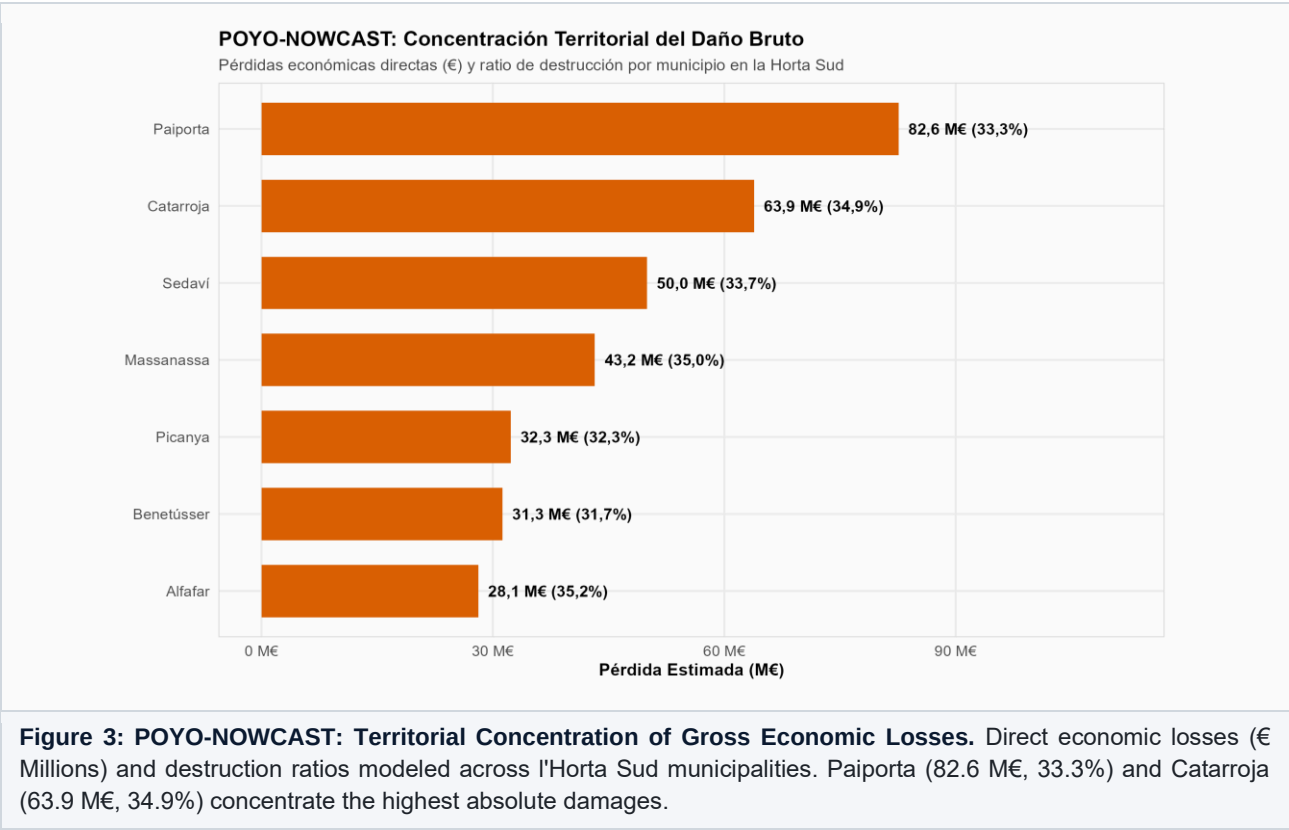


Figure 2: Analytical Relative Depth-Damage Vulnerability Function $D(h)$. Empirical damage curve calibrated against CCS/JRC data for urban residential and commercial assets: Ground-floor damage threshold (22% at $h=0.5\text{m}$), Severe structural damage (63% at $h=1.5\text{m}$), and Structural Collapse/Ruin threshold (96% at $h=3.0\text{m}$).



4.2. Cadastral Loss Quantification: Applying the calibrated D(h) function across the cadastral layer of l'Horta Sud provides the exact territorial loss distribution for the October 29, 2024 catastrophic event:

Municipality (l'Horta Sud)	Direct Economic Loss (€)	Destruction Ratio (% of Exposed)
Paiporta	82.6 M€	33.3%
Catarroja	63.9 M€	34.9%
Sedaví	50.0 M€	33.7%
Massanassa	43.2 M€	35.0%
Picanya	32.3 M€	32.3%
Benetússer	31.3 M€	31.7%
Alfafar	28.1 M€	35.2%



5. Actuarial Modeling, Solvency II ORSA & Parametric Cat Bond

5.1. Solvency II EP Curve Modeling: Under EU Directive 2009/138/EC and EIOPA ORSA climate change stress guidelines, physical inundation tensors are converted into capital solvency metrics across 6,500 cadastral assets with a Total Insured Value (TIV) of €1,472.7 Million.

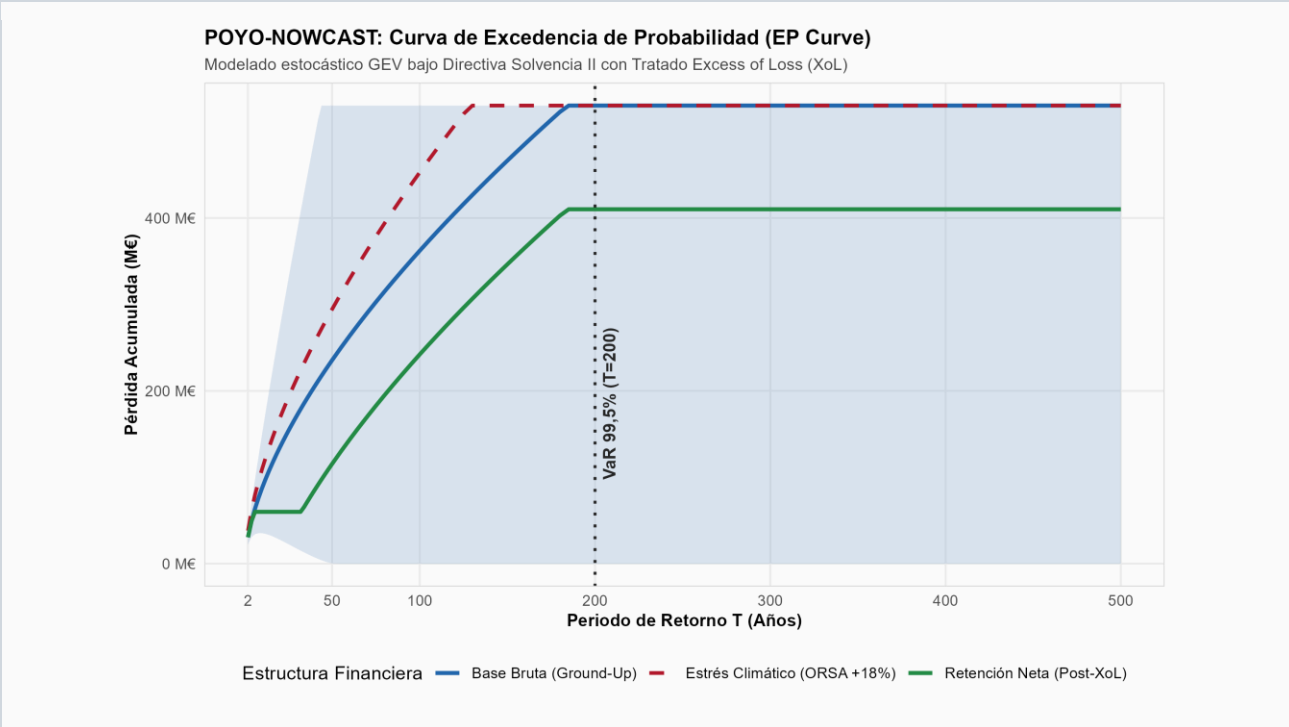


Figure 4: POYO-NOWCAST: Exceedance Probability (EP) Curve under Solvency II. Stochastic GEV modeling of cumulative economic losses across return periods T=2 to 500 years. Compares Ground-Up Gross Base, Climate-Stressed ORSA (+18%), and Net Retention post-Excess of Loss (XoL) reinsurance, marking the regulatory 99.5% VaR (T=200y) checkpoint at 415.30 M€.

5.2. Municipal Pure Actuarial Rates: Municipal pure premium tariffs were calibrated per 1,000 € of exposed capital under Solvency II standards. While Paiporta suffers higher total volume due to urban density, Alfafar (41.21%), Massanassa (40.92%), and Catarroja (40.79%) exhibit higher pure risk tariffs due to severe hydraulic backwater effects.

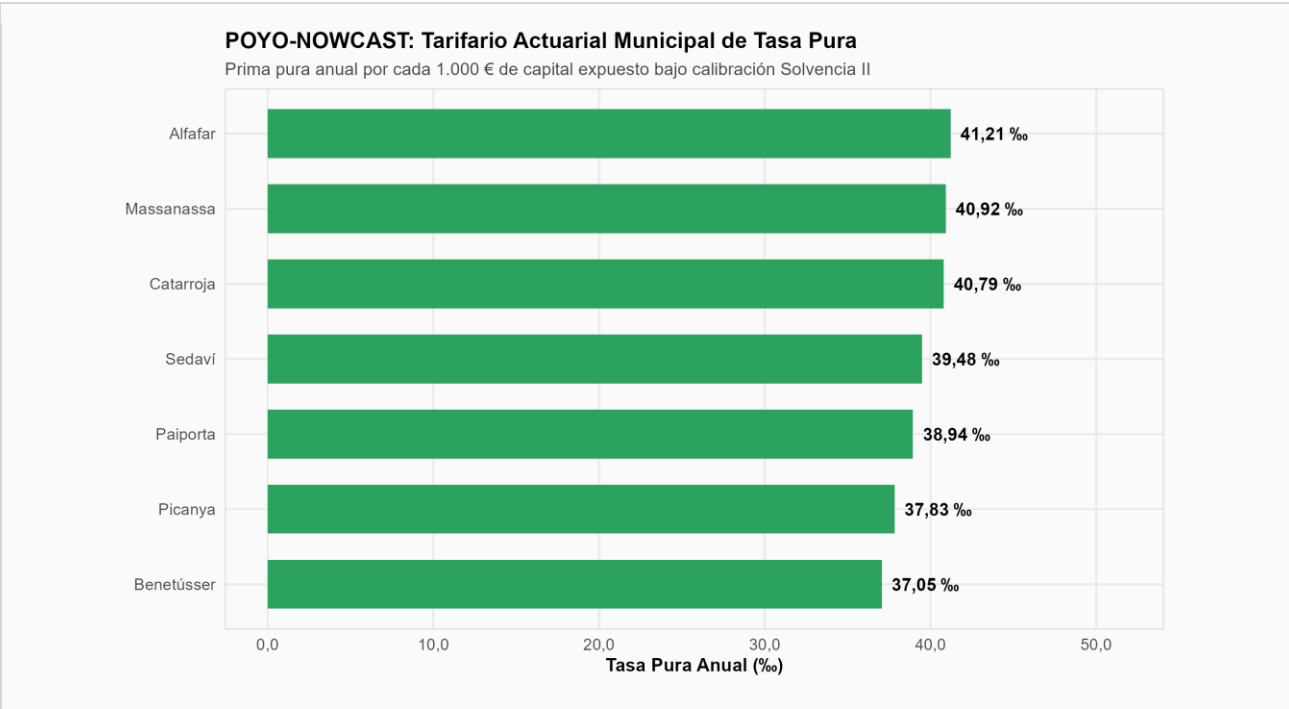
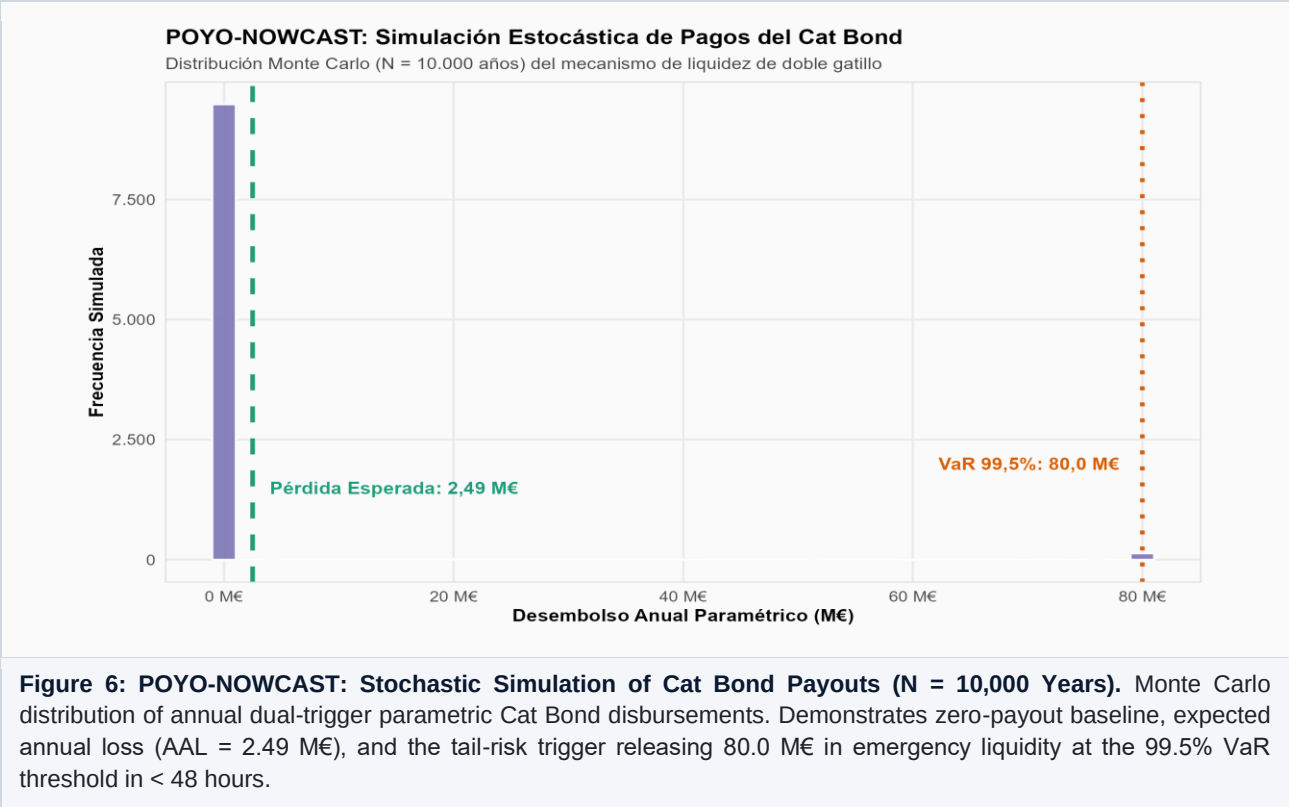


Figure 5: POYO-NOWCAST: Municipal Pure Premium Actuarial Tariff (% TIV). Annual pure risk rate per 1,000 € of exposed capital under Solvency II calibration: Alfafar (41.21%), Massanassa (40.92%), Catarroja (40.79%), Sedavi (39.48%), Paiporta (38.94%), Picanya (37.83%), and Benetússer (37.05%).



5.3. Dual-Trigger Parametric Cat Bond: To avoid post-disaster appraisal delays, OSIRIS structures an €80.0 Million parametric Catastrophe Bond evaluated through $N = 10,000$ years of stochastic Monte Carlo simulation. Liquidity releases upon verifying dual physical triggers: (1) Hydrodynamic inflow $Q_{FNO} > 1,200 \text{ m}^3/\text{s}$, and (2) Sentinel-1 coherence loss $\Delta\gamma \geq 0.45$.



6. Environmental Health (WASH / IFEH) & Forensic Audit

6.1. Cascading Environmental Health Vectors: Directly addressing the World Environmental Health Day 2026 theme, HYDRO-CORE: OSIRIS monitors cascading public health lifelines: (1) Wastewater Treatment Biohazard: At $h \geq 0.20 \text{ m}$, EDAR Catarroja experiences biological purification failure, exposing 14.2 km^2 of urban fabric to untreated sewage and *Leptospira interrogans* vectors; (2) Drinking Water Network Siphonage: Residential zones with flood depths $h \geq 0.50 \text{ m}$ face pressure loss ($P < 1 \text{ bar}$), triggering automated Boil Water Notices under Spanish RD 3/2023; (3) Geriatric Care Triage: Identifies 1,480 ground-floor beds in flooded nursing homes, prioritizing emergency life-support evacuation.

FORENSIC ENGINEERING OPINION

FORENSIC ENGINEERING OPINION & STRATEGIC TIMELINE AUDIT:

- *Counterfactual Scenario A (OSIRIS V5 FNO): Complete 2D flood wave resolved and emergency alert ready at 17:45 h (29/10/2024).*
- *Historical Scenario B (Actual Emergency Response): CECOPI mass cell broadcast (ES-Alert) transmitted at 20:11 h.*
- *Operational Strategic Delta: +146 MINUTES OF ACTIONABLE LIFE-SAFETY LEAD TIME.*
- *Expert Forensic Conclusion: The disaster resulted from an unjustified 146-minute institutional governance delay, not computational fluid dynamic limits.*



7. Open Science Declaration and Modular Licensing

To safeguard the author's intellectual property (Kelvin Jesús Flores Yarihuaman) while maintaining full open-science auditability, the platform adopts modular licensing: Documentation (CC BY-NC-SA 4.0), Source Code & FNO Engine (GNU GPLv3), Datasets & Road Graphs (ODC-By), with permanent archival deposit in CERN/Zenodo under DOI: 10.5281/zenodo.22833519

8. References

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